OMAR LEMKECHER

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EDUCATION

- Bachelor of Science in **Aerospace Engineering** at the **University of California, Los Angeles** Expected Graduation: **2028**

SKILLS

- **Propulsion & Fluids :** N_2O thermodynamics, discharge coefficient modeling, Pressure Vessel design, CFD Basics
- **Structures & Mechanical Analysis :** FEA (ANSYS, SolidWorks), Stresses Analysis (Brackets, Endcaps, Bolt Analysis)
- **Software & Tools :** Python, C++, SolidWorks, Fusion360, Onshape, ANSYS
- **Machining & Prototyping :** CNC machining, CAM, 3D printing, hydrostatic testing, instrumentation integration
- **Interests** : Big Science Fiction reader, Soccer & Volleyball aficionado, Sweet potato fries devourer

EXPERIENCES

Opto-Mechanical Design Intern - TUNSA (Tunisian Space Association) **July-September 2026**

- Designed the optical payload of a **12U CubeSat space telescope**: 200 mm **Cassegrain**, 1028 mm focal length at f/5.14, reaching 0.69″ resolution within a 180 mm envelope
- Performed optical tolerance analysis (**Maréchal λ/14**, Strehl, jitter), setting a **±0.55 µm** secondary mirror alignment budget
- Built an interactive **MATLAB** ray-tracing simulator with live parametric controls and dual-path validation
- Sized **bipod flexure** mirror mounts through bending, buckling, modal and CTE hand calculations, achieving **160×** stress reduction vs straight struts with positive margins on all load cases
- Established the structural load basis per **NASA GEVS** and **CubeSat Design Specification Rev 14**
- Modeled the payload in **SolidWorks** with parametric optical keep-out volumes; traded **6** deployable-structure architectures
- Running **ANSYS FEA** (modal, 20 g static, thermal distortion, random vibration) to verify mount stiffnes

Hybrid Propulsion Feed Systems Lead - Rocket Project at UCLA **2025-Present**

- Lead engineer for the design, analysis and test of Valkyrie's **Feed System**, a **600 lbf N_2O/HTPB SRAD Hybrid Rocket** that successfully launched to **13,300 ft** and fully **successfully recovered**
- Designed full oxidizer feed system (**750 psi MEOP**), including plumbing, fittings interfaces, and pressure relief strategy
- Performed discharge coefficient calculations and flow modeling to characterize pressure losses and transient behavior
- Designed and **FEA validated** lightweight oxidizer tank endcaps and brackets under internal pressure and bolt preload
- Conducted analysis on bolt bearing stresses and bracket bending to meet required margin of safeties
- Optimized tank ullage volume through **test-data-driven iteration** to maximize delivered impulse and combustion stability
- Manufactured all components using both manual and CNC machining across a range of operations

PROJECTS

High-Pressure Vessel Design for Hybrid Rocket Oxidizer Feed System **2025-2026**

- Designed and validated an aluminum **pressure vessel** assembly, 50" cylindrical tank and custom endcaps for a N_2O/HTPB hybrid rocket oxidizer feed system operating at 750 psi MEOP.
- Perfomed analytical **thin wall hoop and longitudinal stress calculations** by hand, then validated by **FEA shell** simulations on **Ansys** to confirm structural margins under full operating pressure applying an industrial margin of safety.
- Conducted a multi-objective design study optimizing simultaneously for pressure load resistance and mass reduction on endcaps and its bolts, applying **Von Mises yield criterion** as the failure condition across iterative FEA runs.
- Full structural **validation** was completed through **hydrostatic pressure testing** of the physical assembly, confirming correlation between analytical predictions, simulation outputs, and real-world load behavior.

66th Parabolic Flight Campaign of the French National Space Studies Center (CNES) **2023**

- Designed and executed a series of **microgravity experiments** covering thermodynamics, centrifugal forces, and Newton's Laws validation. Following a one-year preparation and selection process, two team members flew to **CNES headquarters** in Bordeaux (France) to conduct the experiments aboard a **Zero-G flight.**